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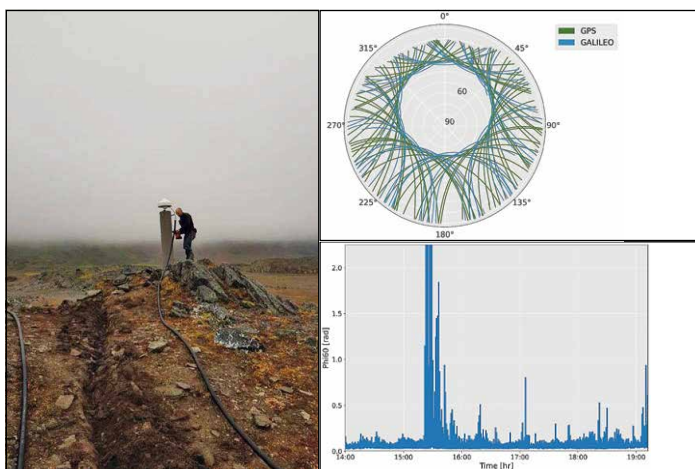
Coordinates

Volume XIX, Issue 9, September 2023

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Testing the **GALILEO HIGH ACCURACY SERVICE** at a high latitude and under ionospheric scintillation

Land registration acceleration in Indonesia



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Cover Image Credit: Hornsund – by Tomasz Wawrzyniak

About Polish Polar Station in Hornsund: The Stanislaw Siedlecki Polish Polar Station in Hornsund is located in the southern part of Spitsbergen island of the Svalbard archipelago in northern Norway. The station was founded in July 1957 by the Polish Academy of Sciences Expedition, and until today it is managed by the Institute of Geophysics of the Polish Academy of Sciences. The year-round Polish Polar Station carries out an extensive monitoring programme, including seismological, Earth magnetism, glaciological and oceanographic observations etc. Its location gives researchers a rare opportunity to study the arctic environment with respect to climate change. (webpage: hornsund.igf.edu.pl)

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Testing the Galileo High Accuracy Service (HAS) at a high latitude and under ionospheric scintillation

In this article, the results obtained assessing the Galileo HAS performance at high latitudes in a period of intense ionospheric activity during the ongoing 25th solar cycle are presented



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Introduction

From the start of 2023, the Galileo High Accuracy Service (HAS) is officially transmitting free Galileo and GPS Precise Point Positioning (PPP) corrections worldwide through the Galileo E6B signal. In addition, HAS corrections are also disseminated via the internet for receivers which do not support E6B.

Already in 2022 valid corrections were transmitted through the Galileo E6B signal for testing purposes. Therefore, since last year an intense research activity has been conducted to assess HAS performance using the live signals/corrections showcasing the effectiveness of the service in terms of dissemination capabilities [1] quality of the corrections [2] and position performance [3-5]. The majority of the work focused on the assessment of the service under nominal conditions and at mid-latitudes. In this work, we analyse the HAS performance at high latitudes during a day characterised by intense ionospheric activity. The article is based on the results presented during the European Navigation Conference (ENC) 2023 [6].

High latitudes can represent a harsh environment for high-accuracy applications due to the reduced visibility of satellites at a high elevation and the occurrence of ionospheric scintillation that can affect the signal quality. Indeed,



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when a radio wave travels through the ionosphere, the presence of irregularities can lead to signal refraction and/or diffraction. These effects can produce phase and amplitude scintillation, namely random fluctuations of the signal phase and/or amplitude. This phenomenon occurs mainly at high latitudes and in the equatorial regions. At high latitudes, the occurrence of scintillation is mainly driven by geomagnetic activity and generally amplitude scintillation is almost absent while phase scintillation can be significant. Phase scintillation can strongly degrade the quality of the signal carrier phase, leading to cycle slips, tracking loss of lock and deteriorating the carrier-based position solutions, such as the ones obtained through PPP [7].

We are currently in the 25th solar cycle, which started in 2019, and going forward the solar cycle maximum, foreseen for 2025/2026. We are also assisting to an enhancement of the ionospheric activity which is also above the forecasts [8]. Therefore, considering the solar activity increase, it is of particular interest to assess the HAS performance under scintillation.

For this purpose, data collected at the Polish Polar Station in Hornsund at the Svalbard are used. The station is equipped with an Ionospheric Scintillation Monitoring Receiver (ISMR) that allows monitoring the occurrence of scintillation events. Moreover, the receiver has been set to record the Galileo E6B navigation message in order to be able to retrieve HAS corrections from the same receiver providing the measurements for the positioning assessment. Therefore, data demodulation is performed in the same challenging conditions. While a first analysis of the HAS demodulation performance at high latitudes was performed in [9], only quiet ionospheric conditions were considered.

In this study, we present an assessment in terms of orbit/clock corrections quality and position performance.

For the analysis, the corrections were extracted using an in-house developed correction parser and then applied to the broadcast data, properly formatted and sent as input to a PPP algorithm to assess the positioning performance. The article is structured as follows. At first, we describe the methodology adopted to demodulate and apply HAS corrections to broadcast data in order to carry out an HAS-based PPP processing. Then the experimental setup is introduced. Experimental results of the PPP analysis are then described. Finally, the conclusions are presented.

Decoding and applying HAS corrections

The HAS corrections, which are transmitted through the live E6B signal, are encoded through a high parity vertical Reed Solomon (RS) code. The HAS message pages are vertically stacked and multiplied by the Reed-Solomon encoding matrix in order to generate a redundant set of pages. Then, each Galileo satellite broadcasts a subset of such pages, which are currently stored in binary format by receivers supporting E6B processing. In order to retrieve the HAS corrections

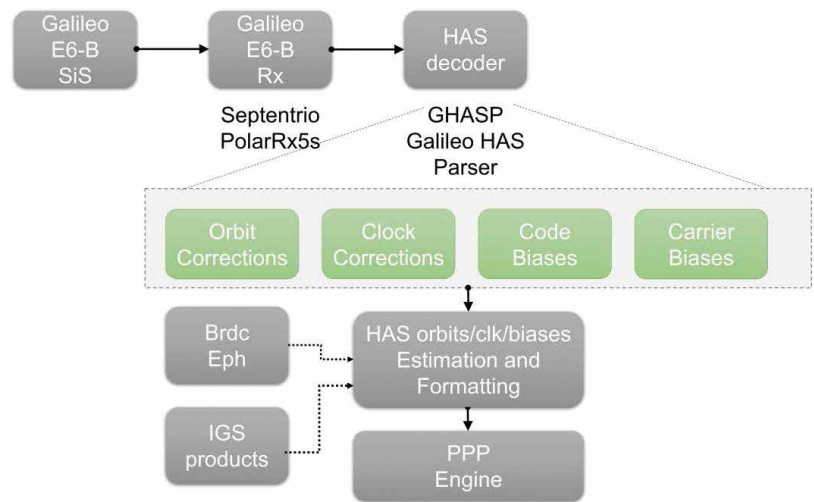


Figure 1: Schematic representation of the process adopted for decoding and applying HAS corrections (extracted from [6])

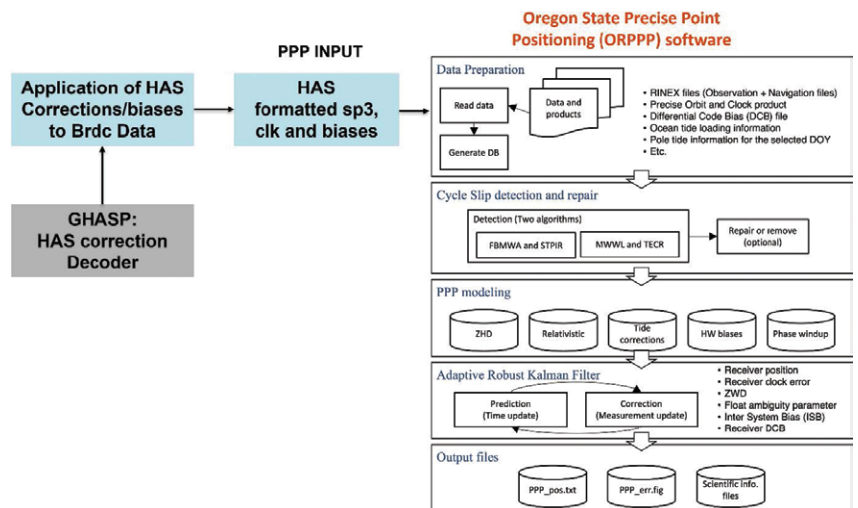


Figure 2: Scheme of the ORPPP software taking as input the HAS formatted sp3, clock and bias files

from the received binary pages, a decoder has been developed in Python [10]. The decoder is now provided as open-source tool and available on github (<https://github.com/borioda/HAS-decoding>).



Figure 3: A researcher from the Warsaw University of Technology installing the antenna of the WUTH reference station

The decoder, denoted as Galileo HAS Parser (GHASP), is able to process binary files from different receiver manufacturers, such as Septentrio, Novatel and Javad. The output of GHASP is a set of Comma-Separated Values (CSV) files containing HAS orbit and clock corrections, code and phase biases. The tool can be used through a user-friendly Graphical User Interface (GUI), which offers plotting options as well. In Figure 1, the full processing chain adopted in this work is illustrated: from the reception of the Galileo E6B signal to the demodulation and application of HAS corrections.

The corrections have been applied to the broadcast data as indicated in [11]. For this purpose, a dedicated script was developed to apply the HAS corrections to the broadcast data and generate Standard Product 3 (SP3)

files that can be easily processed by any commercial and open-source PPP software. In this study, the Oregon State University PPP (ORPPP) engine, developed by the Oregon State University, was used.

The ORPPP software exploits ionosphere-free linear combinations of GNSS measurements and it estimates the unknown vector composed of the position, clock error, zenith wet delay, float ambiguity parameter, differential code biases, and intersystem bias. An adaptive robust Kalman Filter (KF) is used to estimate the unknown parameters [12]. Tidal corrections, phase wind-up, and relativistic corrections are applied as well.

A schematic representation of the processing adopted by ORPPP engine is provided in Figure 2, which also shows the retrieval and integration of the corrections extracted by the GHASP decoder.

Experimental set-up

The data were collected by the WUTH GNSS reference station located in the Polish Polar Station Hornsund (77.00°N, 15.54°E) on Spitsbergen in the Norwegian Svalbard archipelago. Figure 3 shows the the antenna location while an aerial view of the reference station location is provided in Figure 4. Figure 5 shows the map of the Svalbard archipelago indicating the location of the station with the related coordinates.

The WUTH GNSS station is operated by the Warsaw University of Technology and plays an important role in interdisciplinary research like analysis of GNSS signals at high latitudes, monitoring of geodynamic movements (isostatic movements of the Earth's crust), Earth atmosphere monitoring and indirect climate change monitoring (glacier movement, glacier mass balance). Furthermore, the WUTH station is included in international GNSS networks such as the International GNSS Service (IGS), the European Reference Frame (EUREF) Permanent GNSS Network (EPN) and European Plate Observing System (EUPOS).



Figure 4: The Polish Polar Station seen from a drone (picture by Wojciech Flak)



Figure 5: Map view of the WUTH location

The station is equipped with a high-precision reference Septentrio receiver, a PolaRx5S, connected to a SEPCHOKE_B3E6–SPKE choke-ring antenna. The receiver is set to track the full spectrum of current GNSS signals, and it is logging high-frequency (50 Hz) measurements for ionospheric analysis. Indeed, the receiver is an ISMR embedding an ultra-low phase noise Oven Controlled Crystal Oscillator (OCXO). The use of an OCXO ensures that the oscillator phase noise does not exceed the effects of low-level phase scintillation.

The high rate of data sampling and the low noise clock allows for monitoring of the amplitude and phase fluctuations due to ionospheric scintillation. Moreover, for this research, the receiver has been set to record the E6B Galileo navigation messages, in order to be able to demodulate the HAS corrections from the same receiver providing the observations and, consequently, under the same harsh conditions.

Data

The data analysed in this paper were collected from the WUTH station on the 4th of September 2022 (DOY 247). The day

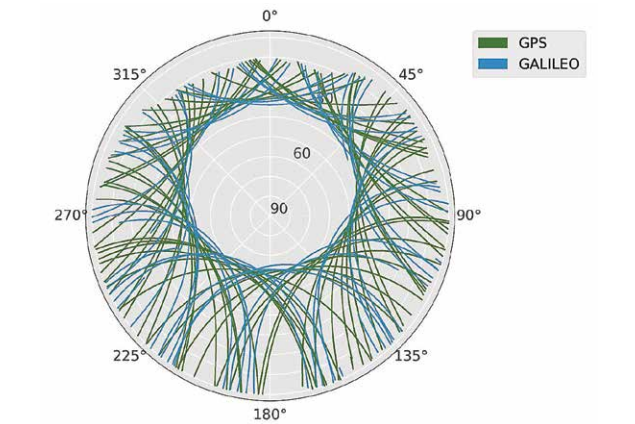


Figure 6: Skyplot seen from the WUTH reference station during the whole day of the 4th September 2022

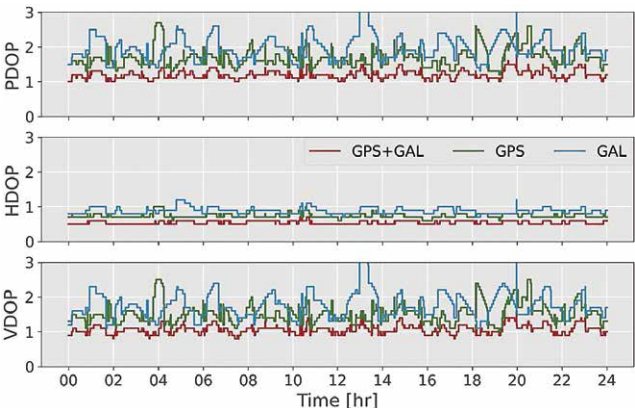


Figure 7: PDOP, HDOP and VDOP evaluated for the WUTH during the full day of the 4th September 2022

was selected for the occurrence of very strong ionospheric activity. In Figure 7 the sky plot observed from the WUTH station for the full selected day is reported. It can be observed that the maximum satellite elevation is around 60 degrees for the entire day. Despite the reduced maximum satellite elevation, good PDOP/HDOP and VDOP values are recorded for the combined GPS + Galileo case (i.e. below 1.5), as shown in Figure 7. In Figure 8 , the level of scintillation is shown through the Phi60 scintillation index reported by the ISMR during a period of the day with strong ionospheric event. As the name suggests, the Phi60 estimates the level scintillation by computing the standard deviation of the detrended carrier phase measurement, averaged over one minute of data [13].

For the analysis, we focused on the interval between 15:17 and 19:00 that was affected by very strong phase scintillation. More specifically, we can observe a very strong phase scintillation peak around 15:20. Moreover, scintillation of weak /moderate intensity is present until the end of the observation period. In the literature slightly different Phi60 thresholds are used to classify different levels of phase scintillation, for example considering different locations. Based on [14] and [15] the values of Phi60 corresponding to different levels of scintillation are reported in Table 1.

Results

The quality of the HAS orbits and clocks was assessed by comparing them with the final precise products downloaded from the Center for Orbit Determination in Europe (CODE). Summary statistics (95th percentile) of the residual errors observed are provided in Table 2 for Galileo and GPS.

The corrections, after proper formatting were then used as input for the PPP processing. PPP processing was set to start just before the very strong scintillation event discussed above. The GPS L1C/A-L2 P and Galileo E1-E5b configuration was selected for the processing. Moreover, the PPP has been configured in static mode. The horizontal and vertical position errors are shown in Figure 9. With respect to the results presented at ENC, in this

Table 1: Phase Scintillation levels adapted by [12] and [13]

Phase Scintillation Level	Phi60 range (radians)
Weak	$0.25 < \text{Phi60} < 0.4$
Moderate	$0.4 < \text{Phi60} < 0.6$
Strong	$0.6 < \text{Phi60} < 0.8$
Very Strong	$\text{Phi60} > 0.8$

Table 2: Orbital/Clock errors (95th percentile)

Constellation	Galileo	GPS
3D Orbital Error	10.5 cm	11 cm
Clock Error	15 cm	11 cm

article, we assess the PPP algorithm by starting the processing even closer to the maximum scintillation peak to further challenge the PPP engine. The processing started at 15:17 only almost 3 minutes before the maximum scintillation peak that occurred at about 15:20. This represents a very challenging and extreme scenario since the PPP convergence starts in the middle of

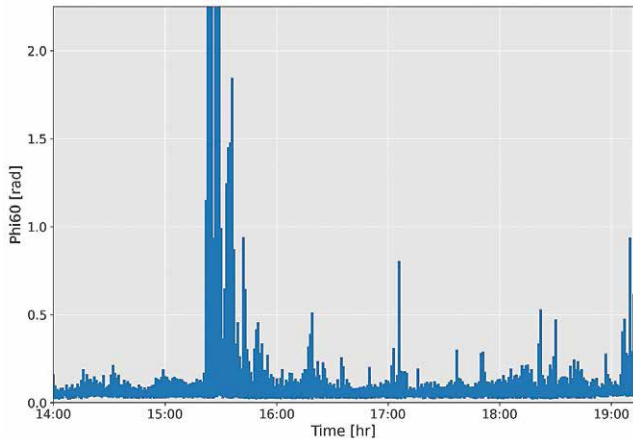


Figure 8: Phi60 recorded for all satellites in view on the L1 band (adapted from [6])

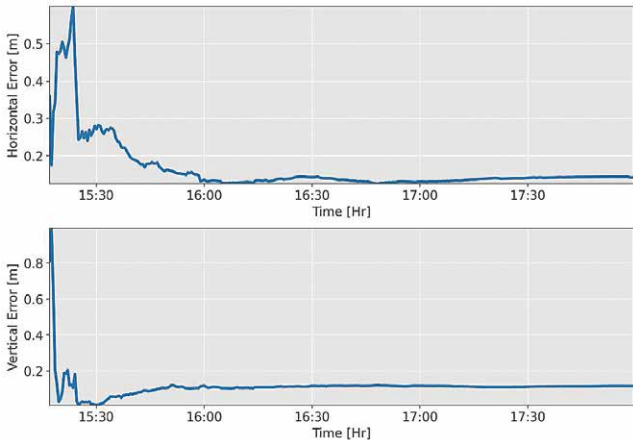


Figure 9: Horizontal and vertical errors under ionospheric scintillation

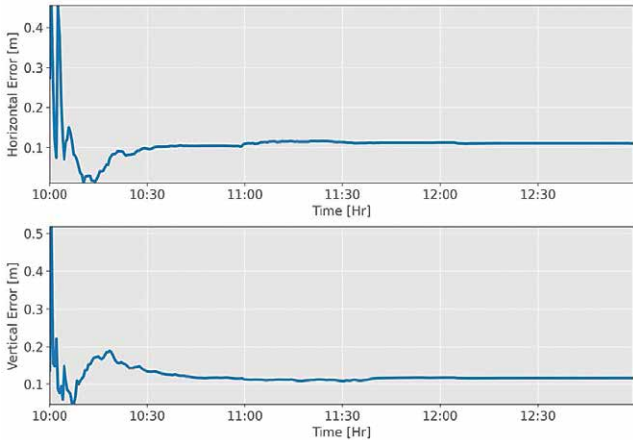


Figure 10: Horizontal and vertical errors under quiet ionospheric conditions

the very strong scintillation period. As expected, this choice impacts the convergence time which is of about 20 minutes. Moreover, it should be underlined that no HAS phase biases have been used, since they were not available yet at the time of the data collection. The engine was set to provide a float PPP solution, so no ambiguity resolution was performed to speed up the convergence. After the convergence the positioning accuracy is well below the maximum target values of 20 and 40 centimetres for the horizontal and vertical cases, respectively.

The error is stable below the target values even if moderate/strong scintillation is still present until the end of the observation period. For comparison, the horizontal and vertical errors are reported in Figure 10 for a period of quiet ionospheric conditions during the same testing day. In this case, the target accuracy is achieved after only 230 seconds (less than four minutes). After convergence, the positioning accuracy is also in this case well below the maximum target values.

Conclusions

Phase scintillation can strongly affect the quality of the signal carrier phase which is crucial for PPP positioning algorithms used by HAS users. In this article, we presented the results obtained assessing the Galileo HAS performance at high latitudes in a period of intense ionospheric activity during the ongoing 25th solar cycle. We analysed a particular day (4th September 2022) when very strong phase scintillation was recorded, and tested PPP convergence using Galileo HAS data applied to GPS and Galileo. The PPP processing was started in the middle of a very strong scintillation event, only few minutes before the maximum scintillation peak. This represents a very extreme event, that as expected, led to an increase of the convergence

High latitudes can represent a harsh environment for high-accuracy applications due to the reduced visibility of satellites at a high elevation and the occurrence of ionospheric scintillation that can affect the signal quality. Indeed, when a radio wave travels through the ionosphere, the presence of irregularities can lead to signal refraction and/or diffraction

time. Despite the very strong phase scintillation event, HAS achieved the target position performance after convergence in a stable way even if scintillation persisted until the end of the observation period. Further work may include a statistical characterization of the convergence time during scintillation (of which only one occurrence is presented here), and using more recent HAS data, particularly after service declaration in January 2023, from which a slightly higher performance is expected.

Acknowledgement

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Reference

- [1] Susi M., Borio D., Gioia C. and Fernandez-Hernandez I. (2022) "Analysis of Different Galileo High Accuracy Service (HAS) Dissemination Options Using Live Signals and Real Corrections," Proceedings of the Navitec Conference, April ESA/ESTEC, Noordwijk, The Netherlands
- [2] Martini I., Susi M., Paonni M., Sgammini M. and Fernandez-Hernandez I. (2022), "Satellite Anomaly Detection with PPP Corrections: A Case Study with Galileo's High Accuracy Service," Proceedings of the International Technical Meeting of The Institute of Navigation, Long Beach, California, January 2022, pp. 1246-1262.
- [3] Naciri N., Yi D., Bisnath S., de Blas J. and Capua R. (2023) "Assessment of Galileo High Accuracy Service (HAS) test signals and preliminary positioning performance". GPS Solutions Vol. 27, No. 73. <https://doi.org/10.1007/s10291-023-01410-y>
- [4] Pintor P., Gonzalez E., Senado A., Bohlig P., Sperl A., Henkel P., Simón J., Hernandez C., de Blas J. and Vazquez J., (2022) "Galileo High Accuracy Service (HAS) Algorithm and Receiver Development and Testing," in Proceedings of the 35th International Technical Meeting of the satellite division of the Institute of Navigation (ION GNSS+ 2022), Denver, Colorado, September 2022, pp. 836-851.
- [5] Fernandez-Hernandez I., Chamorro-Moreno A., Cancela-Díaz S., Calle-Calle J. D., Zoccarato P., Blonski D., Senni T., de Blas J., Hernandez C., Simón J. and Mozo A. (2022) "Galileo High Accuracy Service: Initial Definition and Performance" GPS Solutions, 2022, Vol. 26, No. 65. <https://doi.org/10.1007/s10291-022-01247-x>
- [6] Susi M., Borio D., Azeez A., Park J., Wezka K. and Fernandez-Hernandez I. (2023), "Assessing the High Accuracy Service at High Latitudes", Proc. of the European Navigation Conference (ENC), May-June, Noordwijk, The Netherlands, MDPI Engineering Proceedings
- [7] Aquino M., Dodson, A. Souter, J. Moore, T. (2007) "Ionospheric Scintillation Effects on GPS Carrier Phase Positioning Accuracy at Auroral and Sub-auroral Latitudes". In: Tregoning, P., Rizos, C. (eds) Dynamic Planet. International Association of Geodesy Symposia, vol 130. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-540-49350-1_121
- [8] <https://www.swpc.noaa.gov/products/solar-cycle-progression>
- [9] Susi M., Borio D., Gioia C., Brunes M. T., Dahnn M., Grinde G., Rost C. (2021) "Assessment of Galileo E6-B Data Demodulation Performance at High Latitudes: A Norwegian Vessel Case Study," Remote Sensing 2021, Vol. 13, pp. 46-69, <https://doi.org/10.3390/rs13224669>
- [10] Borio D., Susi M., Gioia C. (2023) "GHASP: A Galileo HAS Parser", GPS Solutions, GPS Toolbox, accepted for publications, July 2023
- [11] European Union (2022) Galileo high accuracy service signal-in-space interface control document (HAS SIS ICD), Issue 1.0, May 2022. https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo_HAS_SIS_ICD_v1.0.pdf
- [12] Guo F. and Zhang X. "Adaptive robust Kalman filtering for precise point positioning", Measurement Science and Technol-ogy, 2014, Vol. 25, No. 10, pp. 1-8, <https://doi.org/10.1088/0957-0233/25/10/105011>
- [13] Van Dierendonck A. J. "Eye on the Ionosphere: Measuring Ionospheric Scintillation Effects from GPS Signals". GPS Solutions 1999 Vol. 2, No. 4, pp. 60– 63. <https://doi.org/10.1007/PL00012769>
- [14] European Union. Galileo High Accuracy Service Signal-in-Space Interface Control Document (HAS SIS ICD). May 2022. Available online: https://www.gsc-europa.eu/sites/default/files/sites/all/files/Galileo_HAS_SIS_ICD_v1.0.pdf
- [15] Dubei S., Wahi R., Mingkhwan E. and Gwal, A.. (2005) "Study of amplitude and phase scintillation at GPS frequency." Indian Journal of Radio and Space Physics. 34.
- [16] Tiwari R., Skone S., Tiwari S., and Strangeways, H.J. "WBM assisted PLL GPS software receiver for mitigating scintillation effects in high latitude region," 2011 URSI General Assembly and Scientific Symposium, Istanbul, Turkey, 2011, pp. 1-4, 